Web Technologies – Project Guidelines

Project Component | Academic Year 2026

Project Completion & Final Evaluation: 15 October 2026

1. Project Overview

Students will work in **batches of 3–4** and develop a web-based software project applying the concepts and technologies covered in the Web Technologies course.

> **Project Duration:** 13 September 2026 – 15 October 2026
> **Project Component:** 12 hours
> **Team Size:** 3–4 students
> **Project Assessment:** 25 marks

2. General Project Requirements

Each project should:

- Address a clearly defined real-world problem or requirement.
- Have clearly defined objectives and functional requirements.
- Provide a working web-based application.
- Use **Next.js** for the frontend.
- Use **NestJS** for backend development.
- Use **GraphQL** for API communication.
- Include appropriate database/cloud connectivity.

- Implement suitable validation and error handling.
- Implement authentication and authorization wherever required.
- Use Git/GitHub for version control and collaborative development.
- Demonstrate Docker-based containerization.
- Demonstrate a basic CI/CD workflow.
- Demonstrate Kubernetes/Minikube deployment.
- Demonstrate basic monitoring, logging, or error tracking.

3. Suggested Technology Stack

Component	Technology / Requirement
Frontend	Next.js
UI / Styling	Tailwind CSS or suitable CSS approach
Backend	NestJS
API	GraphQL
Database	Suitable relational / NoSQL / cloud database
Version Control	Git + GitHub
Containerization	Docker
CI/CD	GitHub Actions or equivalent
Deployment	Kubernetes / Minikube
Monitoring / Error Tracking	Sentry / Prometheus / Grafana or equivalent

4. Project Schedule

Phase	Activity	Date	Hours	Assessment
Phase 1	Preliminary Design of the Project	15 September 2026	2	—
Phase 2	Zeroth Presentation	19 September 2026	2	—
Phase 3	Project Work – First Phase	26 September 2026	2	**Evaluation 1 – 10 Marks**
Phase 4	Interim Presentation	3 October 2026	2	—
Phase 5	Project Work – Final Phase & Report Writing	10 October 2026	2	**Report – 5 Marks**
Phase 6	Final Evaluation, Presentation and Exhibition	15 October 2026	2	**Evaluation 2 – 10 Marks**

5. Phase 1 – Preliminary Design of the Project

Date: 15 September 2026 | **Duration:** 2 Hours

Students should finalize

- Project title
- Problem statement
- Objectives
- Target users / stakeholders
- Major functional requirements
- Proposed solution

- Proposed technology stack
- Basic system architecture
- Major modules/features
- Team member responsibilities

Slides to Include

1. Title Slide – Project title and team members
2. Problem Statement
3. Objectives
4. Target Users / Stakeholders
5. Existing Problem / Motivation
6. Proposed Solution
7. Major Features / Modules
8. Proposed Technology Stack
9. High-Level System Architecture
10. Team Member Roles

> **Expected Output:** A clearly defined project idea and preliminary design.

6. Phase 2 – Zeroth Presentation

Date: 19 September 2026 | **Duration:** 2 Hours

Purpose

Present the finalized project proposal and obtain feedback before implementation.

Slides to Include

1. Title Slide
2. Problem Statement
3. Objectives
4. Target Users / Stakeholders

5. Existing System / Current Challenges
6. Proposed System
7. Functional Requirements
8. Major Modules
9. System Architecture
10. Technology Stack
11. Database / Data Model – preliminary design
12. UI Wireframes / Initial Screens
13. Team Responsibilities
14. Development Plan / Milestones

> **Expected Output:** Approved project direction, requirements and design.

7. Phase 3 – Project Work: First Phase

Date: 26 September 2026 | **Duration:** 2 Hours

Evaluation 1 – 10 Marks

Students should demonstrate

- Project setup and repository
- Initial Next.js frontend
- Initial NestJS backend
- Database setup
- Initial GraphQL schema/API
- Core application functionality
- Initial frontend-backend integration
- Progress against the approved requirements

Slides to Include

1. Project Overview
2. Problem Statement and Objectives

3. Finalized Requirements
4. System Architecture
5. Technology Stack
6. Implemented Modules / Features
7. Frontend Screenshots / Demo
8. Backend / GraphQL Implementation
9. Database Design / Implementation
10. Current Project Status
11. Individual Contribution of Team Members
12. Challenges Faced and Solutions
13. Plan for Remaining Development

> **Expected Output:** A working prototype demonstrating the first phase of implementation.

8. Phase 4 – Interim Presentation

Date: 3 October 2026 | **Duration:** 2 Hours

Purpose

Review the progress of the project and identify the remaining work before final development.

Slides to Include

1. Title Slide
2. Project Objectives
3. System Architecture
4. Completed Features
5. Working Application Screens
6. Frontend Implementation
7. Backend / GraphQL Implementation

8. Database / Cloud Integration
9. Authentication / Authorization, if applicable
10. Docker / Containerization Progress
11. CI/CD Progress
12. Deployment Progress
13. Testing Status
14. Problems / Challenges
15. Remaining Features and Work Plan

> **Expected Output:** Substantially working application with clear identification of remaining work.

9. Phase 5 – Final Phase & Report Writing

Date: 10 October 2026 | **Duration:** 2 Hours

Report – 5 Marks

Students should complete

- All major functional requirements
- Frontend-backend integration
- Database integration
- Testing and debugging
- Dockerization
- CI/CD workflow
- Kubernetes/Minikube deployment
- Monitoring / error tracking
- Final project report

Report should Contain

1. Title Page
2. Abstract

3. Introduction
4. Problem Statement
5. Objectives
6. Requirements
7. System Architecture
8. Technology Stack
9. System Design / Database Design
10. Implementation
11. API / GraphQL Details
12. Testing
13. Deployment
14. Results / Screenshots
15. Conclusion
16. Future Scope
17. References

> **Expected Output:** Complete project ready for final evaluation and exhibition.

10. Phase 6 – Final Evaluation, Presentation & Exhibition

Date: 15 October 2026 | **Duration:** 2 Hours

Evaluation 2 – 10 Marks

Final Presentation Slides

1. Title Slide
2. Problem Statement
3. Objectives
4. Proposed Solution
5. System Architecture
6. Technology Stack

7. Major Features
8. UI Screens / Working Application
9. Backend and GraphQL Implementation
10. Database / Cloud Integration
11. Authentication / Authorization
12. Dockerization
13. CI/CD Pipeline
14. Kubernetes Deployment
15. Monitoring / Error Tracking
16. Testing and Results
17. Challenges and Solutions
18. Individual Contributions
19. Conclusion and Future Scope

Final Demonstration

- Demonstrate the complete working application.
- Demonstrate the major user workflows.
- Demonstrate GraphQL queries/mutations.
- Demonstrate database operations.
- Demonstrate authentication/authorization where applicable.
- Demonstrate Docker deployment.
- Demonstrate CI/CD workflow.
- Demonstrate Kubernetes deployment.
- Demonstrate monitoring/error tracking.

> **Expected Output:** Complete working project, final presentation and exhibition-ready demonstration.

11. Assessment Structure

Component	Marks	Recommended Assessment Point
Evaluation 1	**10**	Phase 3 – Project Work: First Phase
Evaluation 2	**10**	Phase 6 – Final Evaluation
Project Report	**5**	Phase 5 / Final Submission
Total	**25**	

12. General Instructions

- Each team must consist of **3–4 students**.
- Each team should select a suitable and preferably unique project topic.
- The project must demonstrate meaningful application of the course concepts.
- All team members must contribute to the development.
- Use GitHub for version control and maintain regular commits.
- Students must understand the code they submit.
- Use of external libraries, frameworks and APIs is permitted where appropriate, but students must be able to explain their use.
- Projects copied substantially from existing sources will not be accepted.
- Students should maintain regular progress between the scheduled project phases.
- Each student must be able to explain their individual contribution during evaluation.
- The final application should be functional and demonstrable.
- All required presentations should follow the slide contents specified for the respective phase.
- The final report should be submitted before the final evaluation.

> **Important:** The six project phases and their 2-hour allocation follow the project structure specified in the course syllabus. The assessment mapping shown here places Evaluation 1 after the first implementation phase and Evaluation 2 at the final evaluation, with 5 marks for the report.

13. Final Deadline

15 October 2026

Final Evaluation, Presentation, Exhibition and Project Submission